

Computational Biology Assignment 2

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Viterbi path

None = -1
Exon = 0
Intron = 1

Patient Alpha

path: [0, 0, 0, 0]

path_prob: 0.00032036132812500006

delta (states x observation):

	t1	t2	t3	t4	t5
E	0.125	0.028125	0.006328125	0.0014238281	0.00032036133
I	0.2	0.008	0.00096	3.84e-05	2.1357422e-05

psi (states x observation):

	t1	t2	t3	t4	t5
E	-1	0	0	0	0
I	-1	1	1	1	0

Patient Beta

path: [1, 1, 1, 1]

path_prob: 0.0020971520000000001

delta (states x observation):

	t1	t2	t3	t4	t5
E	0.125	0.028125	0.006328125	0.0014238281	0.00032768
I	0.2	0.064	0.02048	0.0065536	0.002097152

psi (states x observation):

	t1	t2	t3	t4	t5
E	-1	0	0	0	1
I	-1	1	1	1	1

Baseline ODE

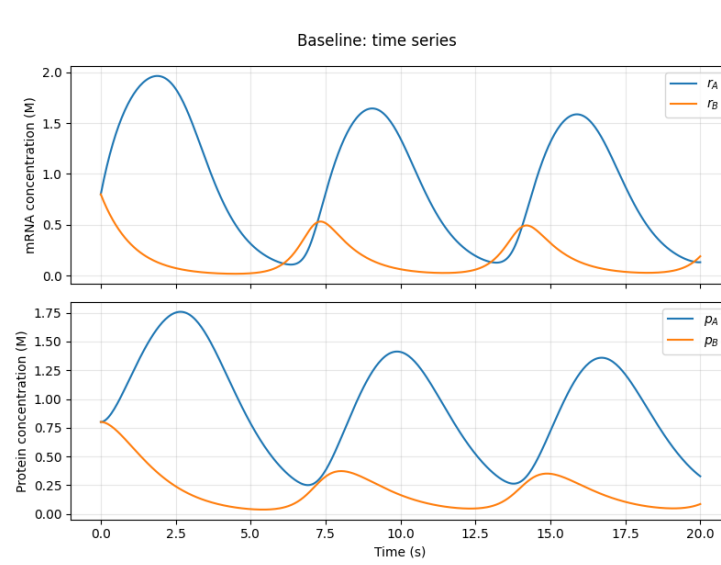


Figure 1:

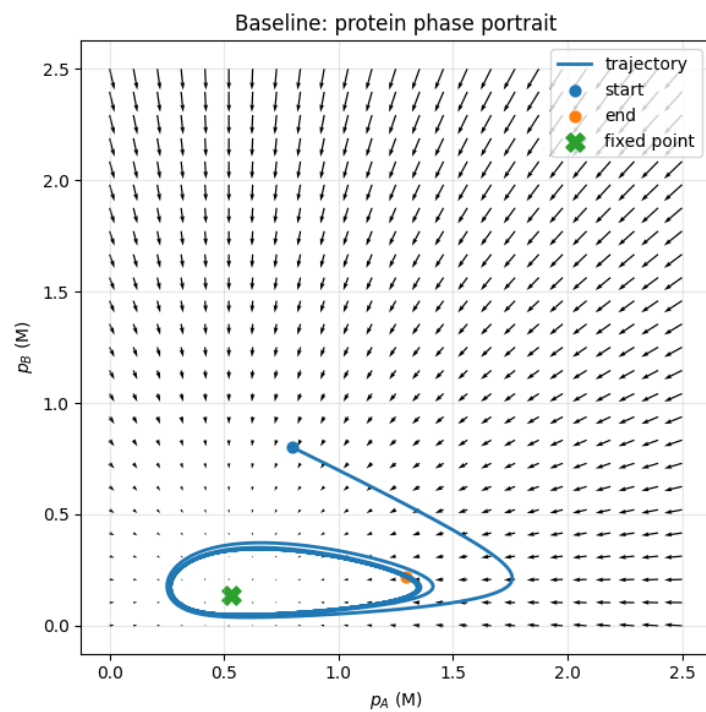


Figure 2:

Patient Alpha

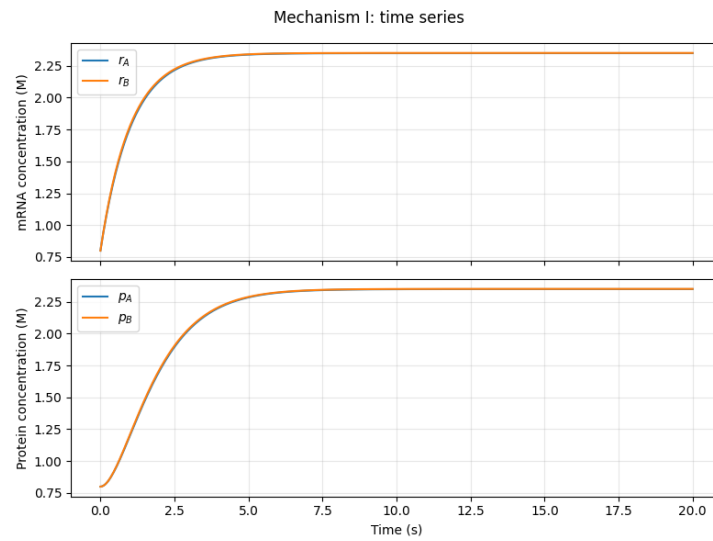


Figure 3:

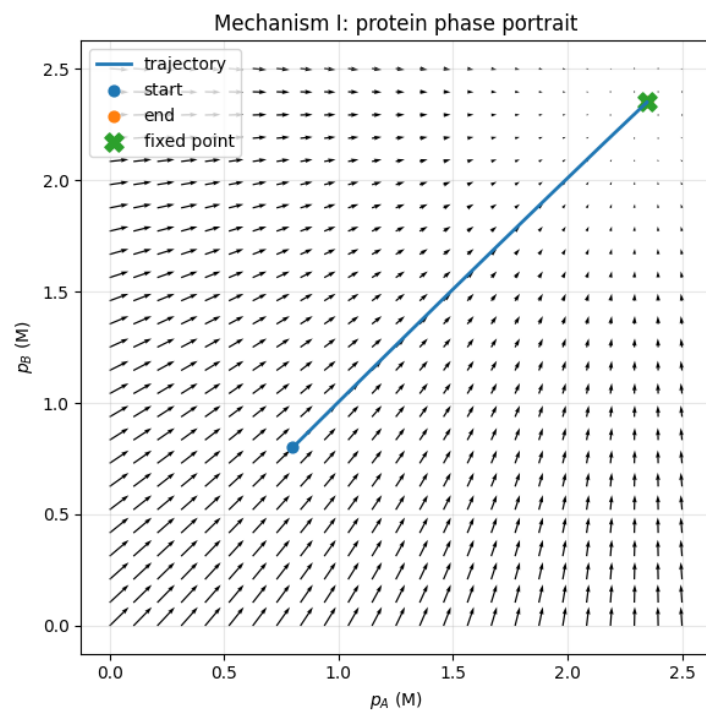


Figure 4:

Patient Beta

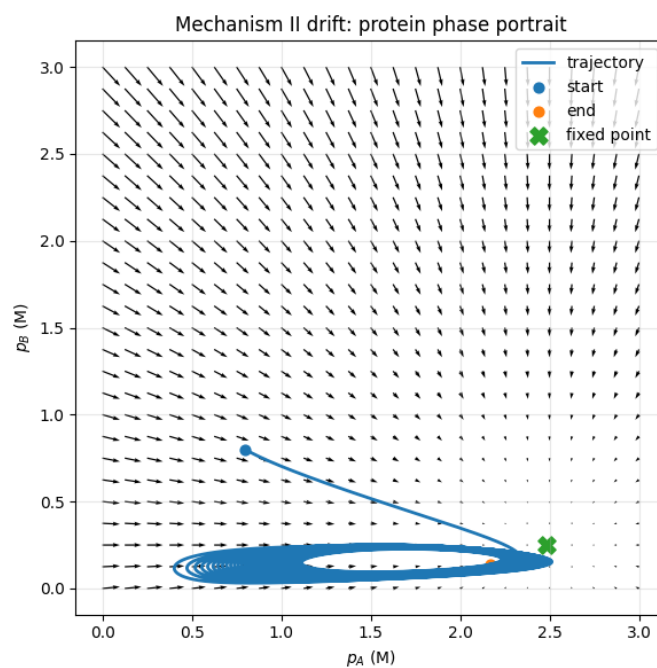


Figure 5:

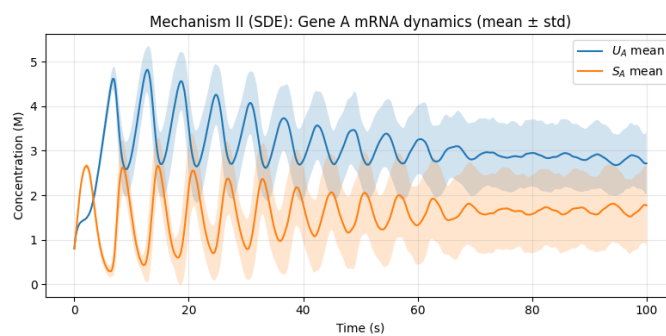


Figure 6:

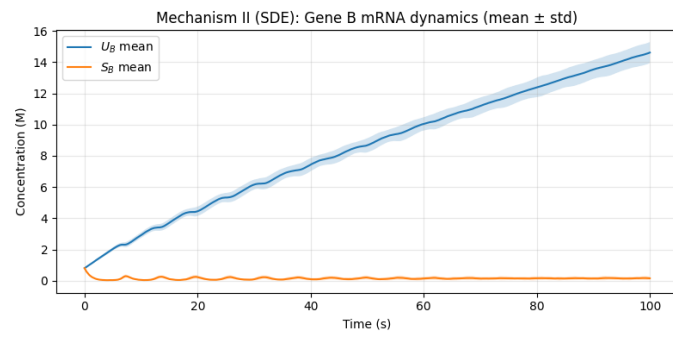


Figure 7:

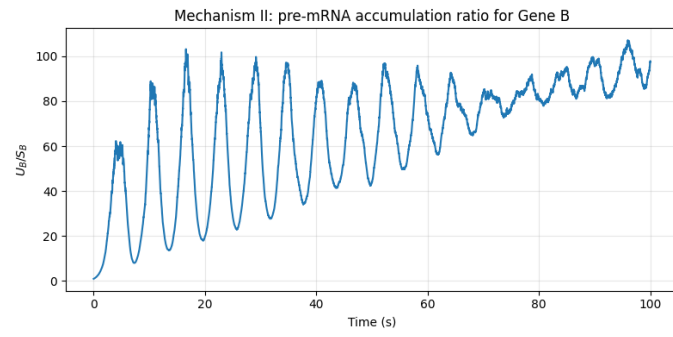


Figure 8:

Bonus

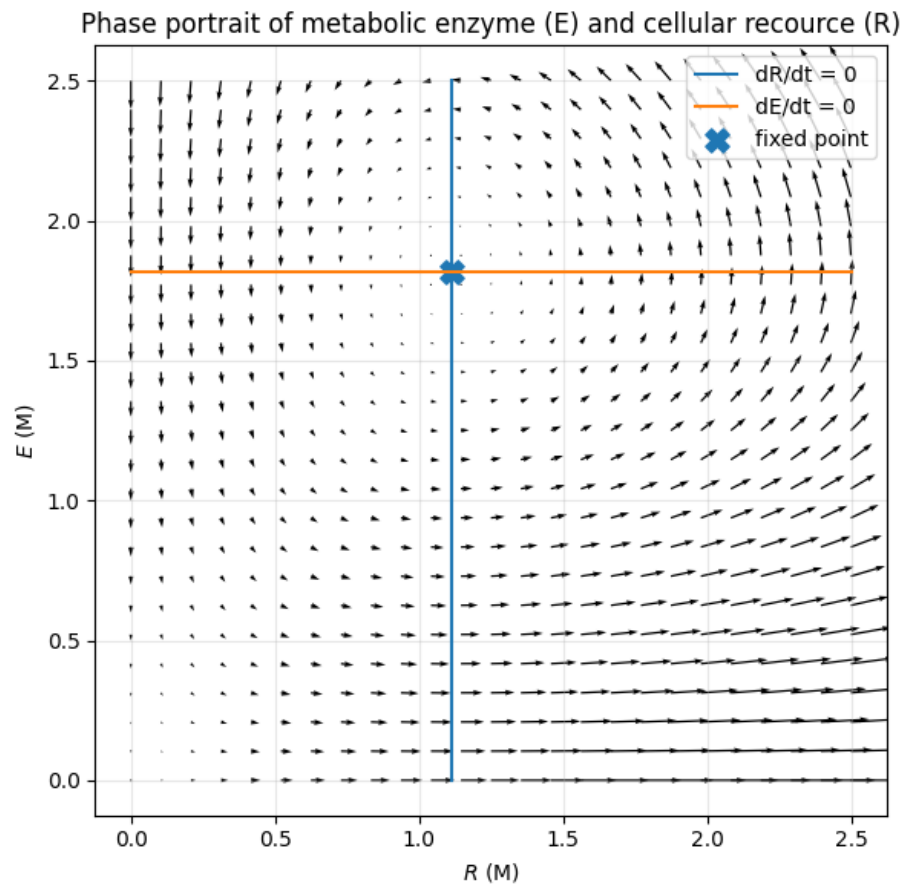


Figure 9: